Data Science

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Outline

- Introduction to data
- What is data science?
- What is data science not?
- (A few) data science examples
- Course objectives and topics
- Course logistics

Introduction to data

- **What are qualitative and quantitative data?**
- **Difference between discrete and continuous data**
- **Real world applications of data**

What is data?

- We often use the term data to refer to computer information
- This information is either transmitted or stored
- Data comes in numerous forms
- Any kind of information may it be in numbers or text, or pictures is termed as Data

Types of data

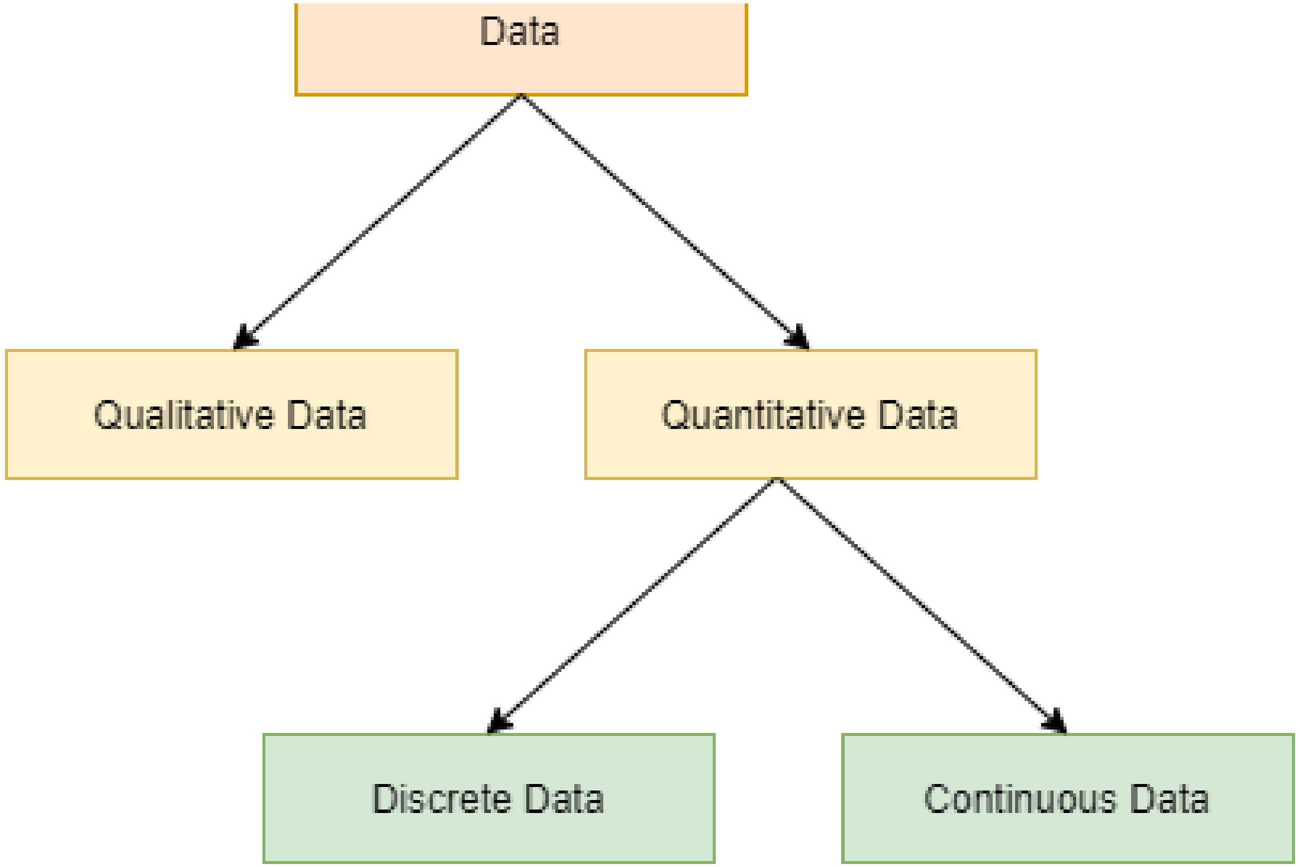
- Data comes in different types. Some of the common types of data include:
  - Text
  - Image
  - Video
  - Numbers
  - Spreadsheets
  - Sound

Qualitative vs Quantitative data

Qualitative	Quantitative
• Qualitative data is the data that is a descriptive piece of information. • Deals with descriptions • Data can be observed not measured • For example, "What a nice day it is"	• Quantitative Data is the data that is numerical information • Deals with numbers • Data can be measured • For example, “1”, “3.65” etc.

Qualitative vs Quantitative data

Quantitative data can be of two types



Discrete vs Continuous data

Discrete	Continuous
• Can be expressed as a specific value. • For example, “Number of months in a year“, “Number of members in a family” etc.	• Can be any value in an interval • For example, “The amount of oxygen in the atmosphere”, “Age of members in a family”

Real world applications of data

Predicting interests of the audience on different online video streaming platforms

Getting insights from customer reviews in online stores, food delivery apps etc.

Effective targeting of the advertisements

What is data science?

- **Some possible definitions:-**
- **Data science** is the application of computational and statistical techniques to address or gain insight into some problem in the real world

Extracting meaningful insights from data is known as data science.

Data when investigated and carefully analyzed, provides insights which enriches our daily lives

What is data science?

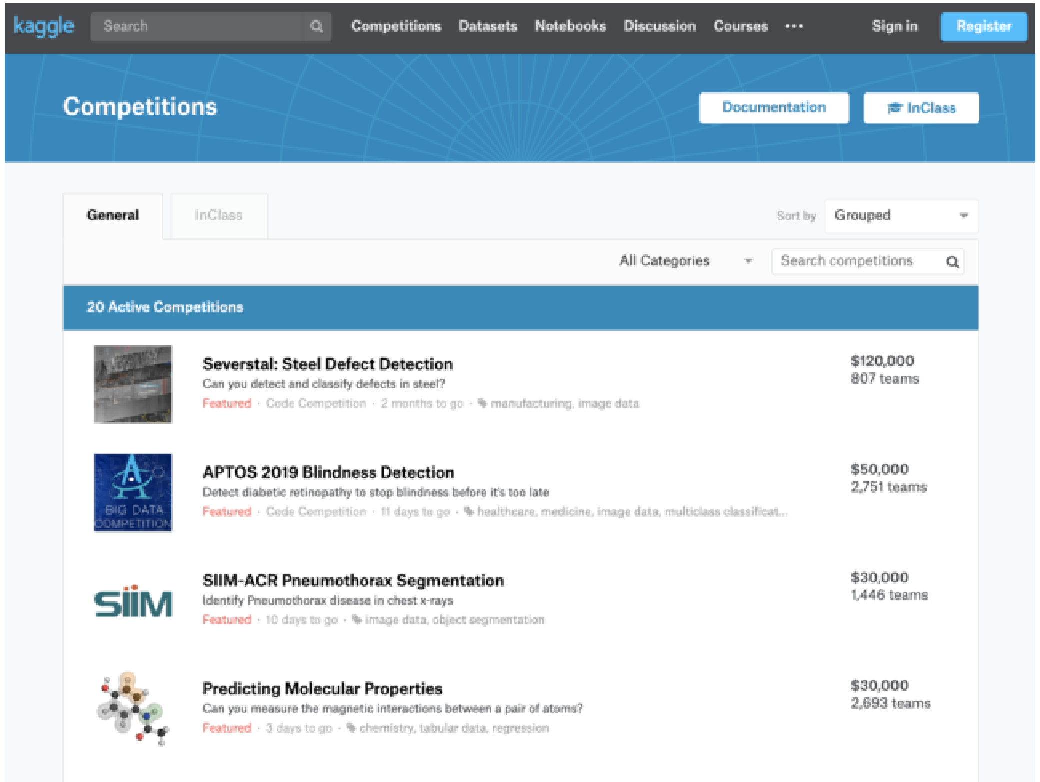
- **Data science** = statistics +
  data processing +
  machine learning +
  scientific inquiry +
  visualization +
  business analytics +
  big data + …

Data science is not machine learning

- Machine learning involves computation and statistics, but has not (traditionally) been very concerned about answering scientific questions
  - Machine learning has a heavy focus on fancy algorithms…

- ... but sometimes the best way to solve a problem is just by visualizing the data, for instance

Data science is not machine learning competitions



- Data science competitions like Kaggle ask you to optimize a metric on a fixed data set
- This may or may not ultimately solve the desired business/scientific problem
- Data science is the iterative cycle of designing a concrete problem, building an algorithm to solve it (or determining that this is not possible), and evaluating what insights this provides for the real underlying question

Data science is not statistics

- “Analyzing data computationally, to understand some phenomenon in the real world, you say? … that sounds an awful lot like statistics”
- Statistics (at least the academic type) has evolved a lot more along the mathematical/theoretical frontier Not many statistics courses have a lecture on e.g. web scraping, or a lot of data processing more generally

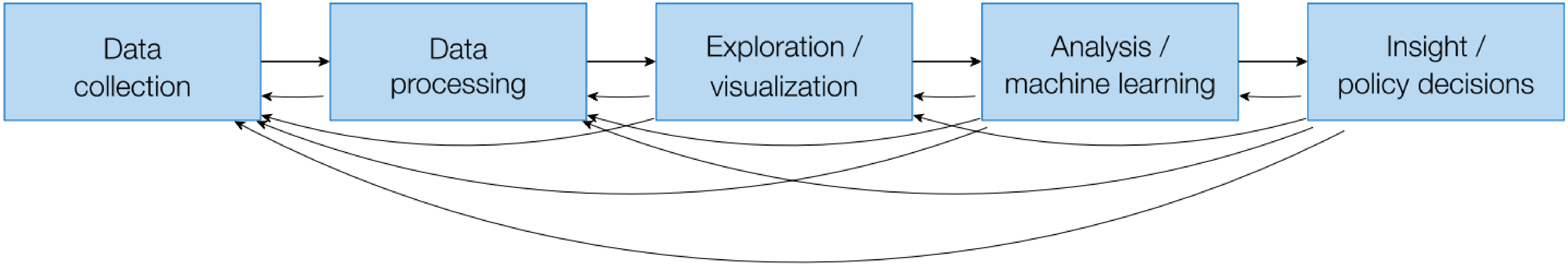
- Plus, statisticians use R, while data scientists use Python ... clearly these are completely different fields

Data science is not big data

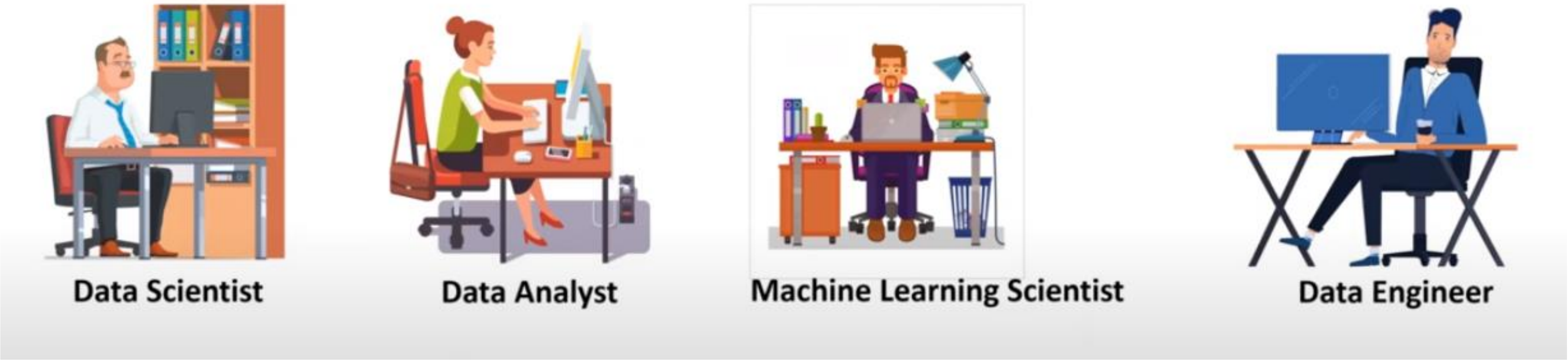
- Sometimes, in order to truly understand and answer your question, you need massive amounts of data…
  - …But sometimes you don't

- Don't create more work for yourself than you need to

Back to what data science is

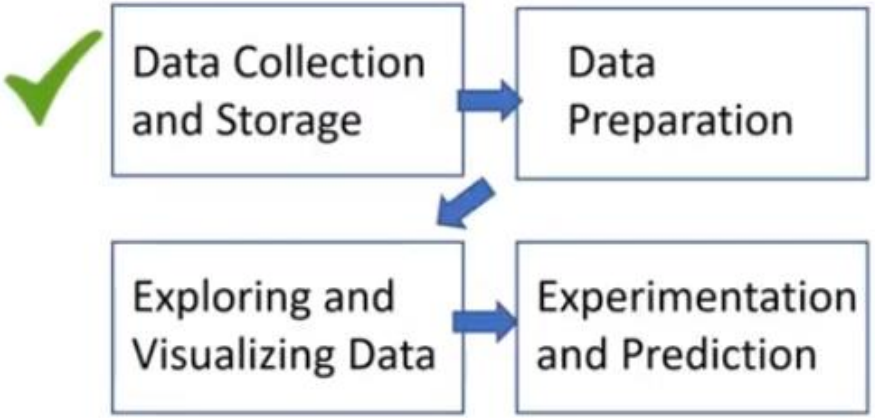


Careers in data science?



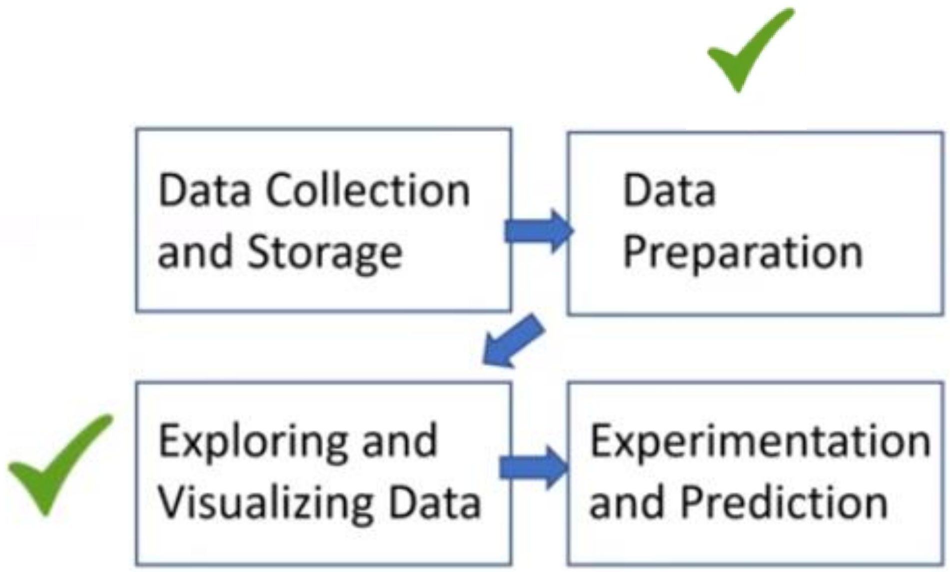
Data Engineer

- **Data engineer**:
  - Information architect
  - Build data pipelines and storage solutions
  - Maintain data access



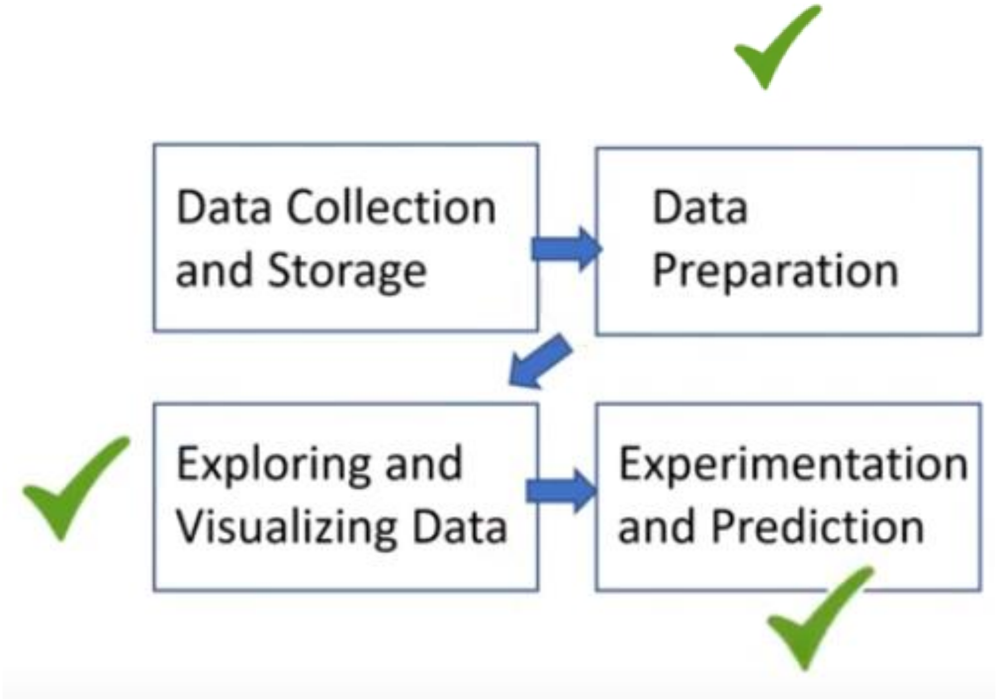
Data Analyst

- **Data analyst**:
  - Perform simpler analyses that describe data
  - Reports and dashboard to summarize data
  - Clean data for analysis



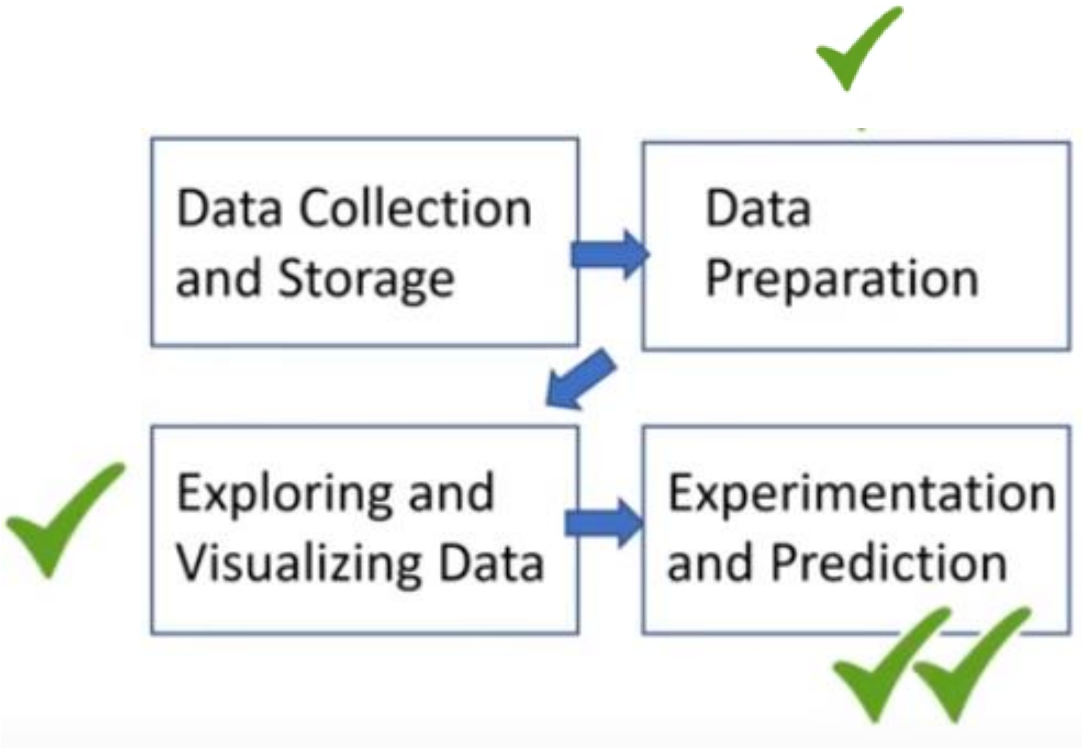
Data Scientist

- Versed in statistical methods
- Run experiments and analyses for insights
- Traditional machine learning



Machine Learning Scientist

- Predictions
- Classification
- Deep Learning



How data science helps us?

- Simply stated, data science helps us answer different types of questions from data. Some common questions to ask from data are:
  - Which class does this belong to - A or B?
  - Is this an outlier?
  - What will probably be the value of this variable?
  - What should be done now?

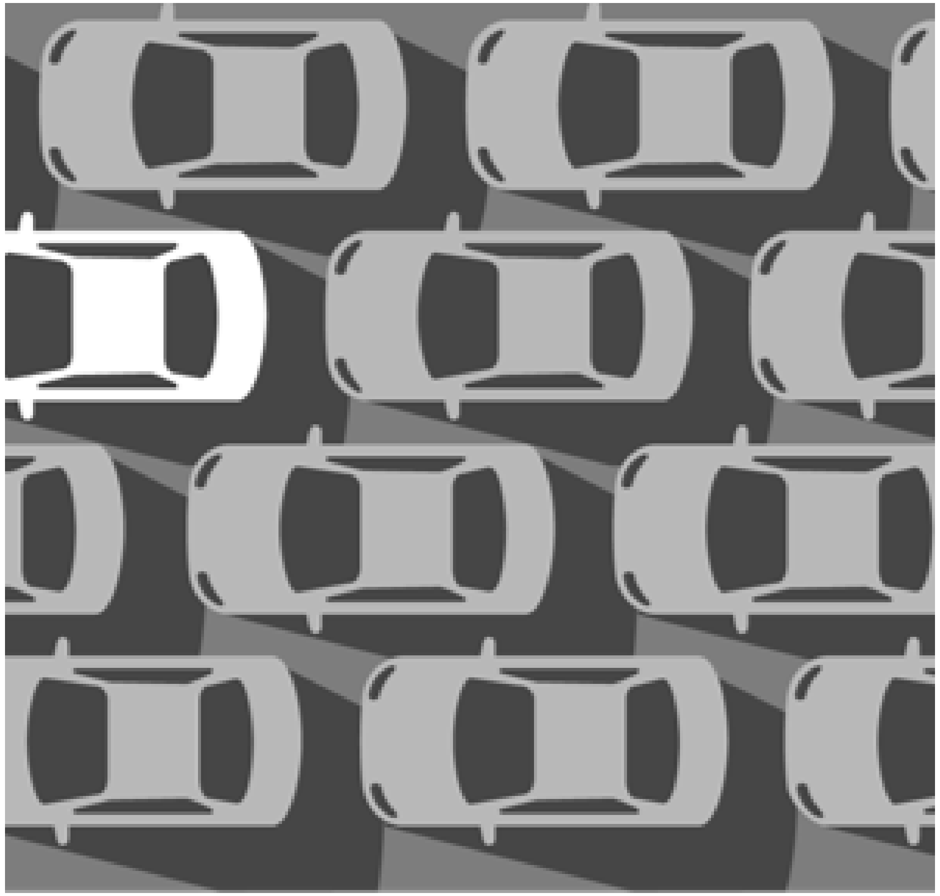
Which class does this belong to - A or B?

- The answers to some questions can only be from a definite number of options.
- For example,
- Q: Will it rain today?
  - A: Yes/No
- Q: Will the weather be hot or cold?
  - A: Hot/Cold

- To make such predictions, we use a family of algorithms called classification algorithms.

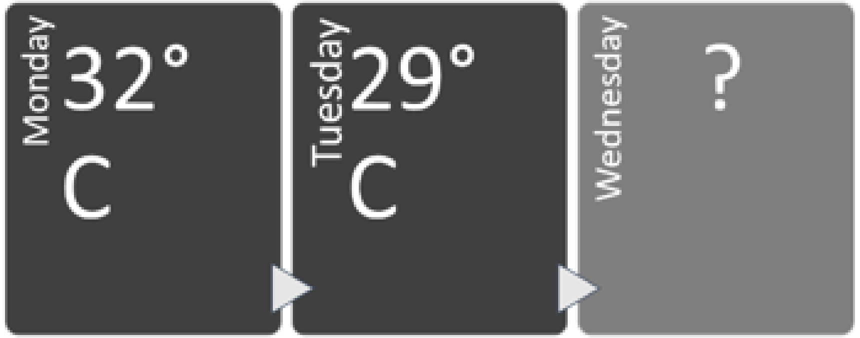
Is this an outlier?

- In some cases, the objective is to find outliers or anomalies in data that is otherwise mostly consistent. Some examples of anomaly detections are:
- Q: Is this email normal or spam?
- Q: You are checking your car tyre pressure. Is the reading normal?
- The algorithms that are used for these types of questions are called anomaly detection algorithms.



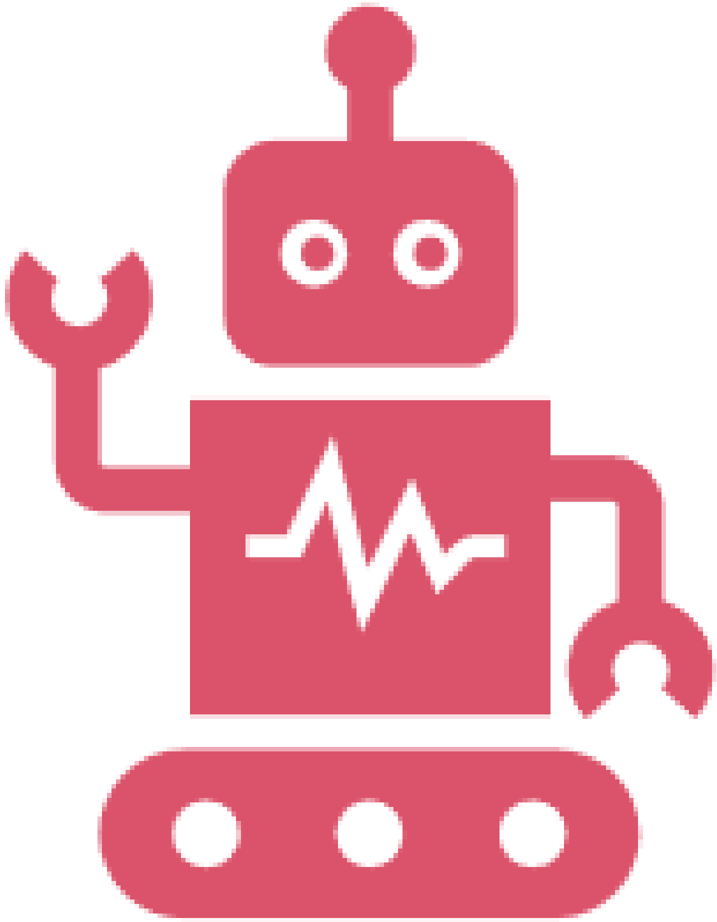
What will probably be the value of this variable?

- There are scenarios in which we must predict numerical values of a variable based on historic data. Some examples are:
- Q: How much rainfall will we receive this year?
- A: 100 mm
- Q: How many runs will the winning team score?
- A: 320
- The kind of algorithms that can predict these values are called regression algorithms.



What should be done now?

- This question usually solves the problems of autonomous robots or self-driving cars that need to make decisions based on changes in external factors. Machine learning helps to solve such problems with the help of reinforcement learning.



Learning objectives of this course

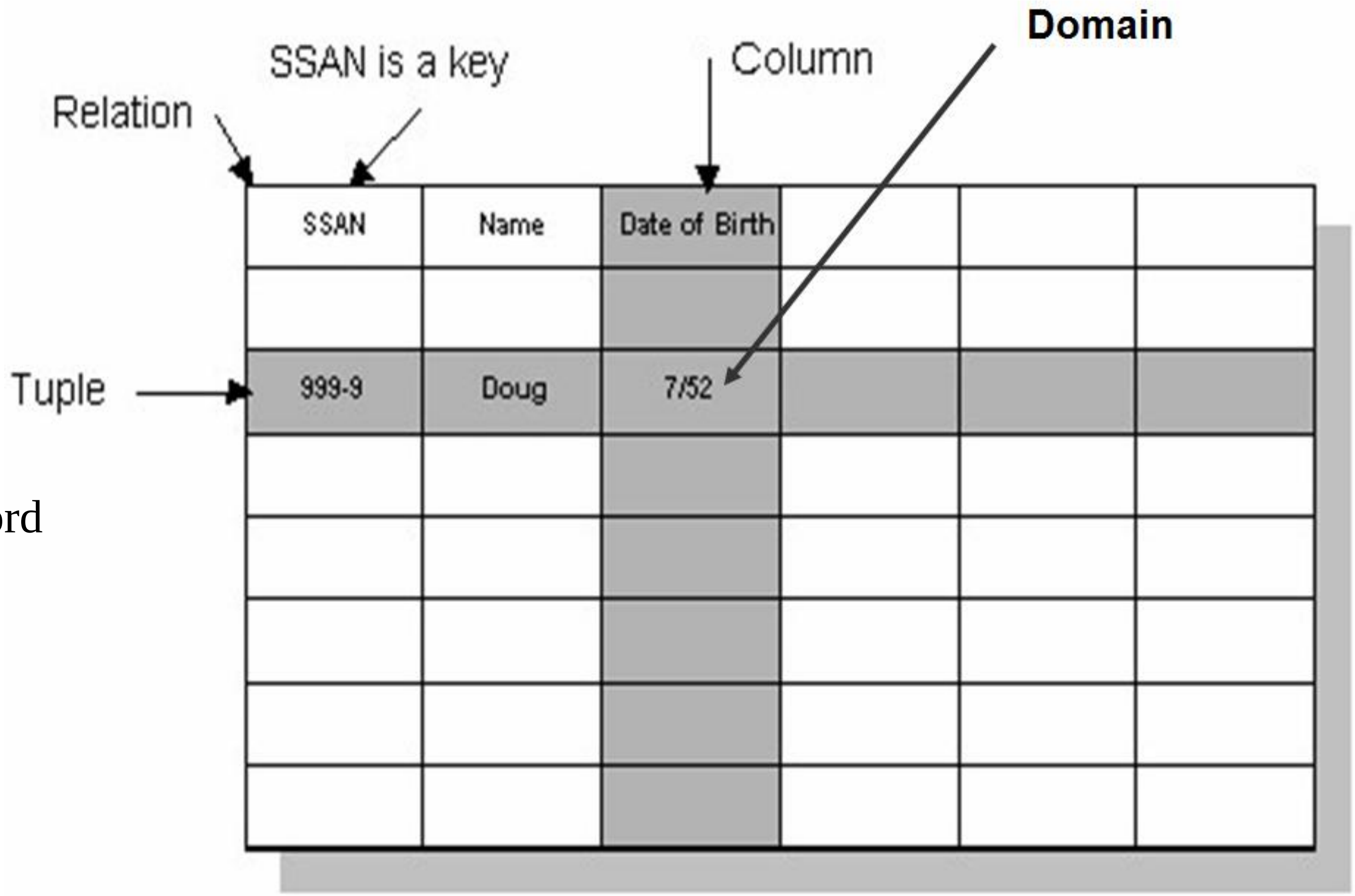
- After taking this course, you should…
- … understand the full data science pipeline, and be familiar with programming tools to accomplish the different portions
- ... be able to collect data from unstructured sources and store it using appropriate structure such as relational databases, graphs, matrices, etc
- ... know to explore and visualize your data
- ... be able to analyze your data rigorously using a variety of statistical and machine learning approaches

Topics covered

- **Data collection** and management: relational data, matrices and vectors, graphs
- and networks, free text processing, geographical data
- **Statistical modeling and machine learning:** linear and nonlinear classification
- and regression, regularization, data cleaning, hypothesis testing, kernel methods
- and SVMs, boosting, clustering, dimensionality reduction, recommender systems,
- deep learning, probabilistic models, scalable ML
- **Visualization:** basic visualization and data exploration, data presentation and
- interactivity

Relational Database

Column/Field/Attribute



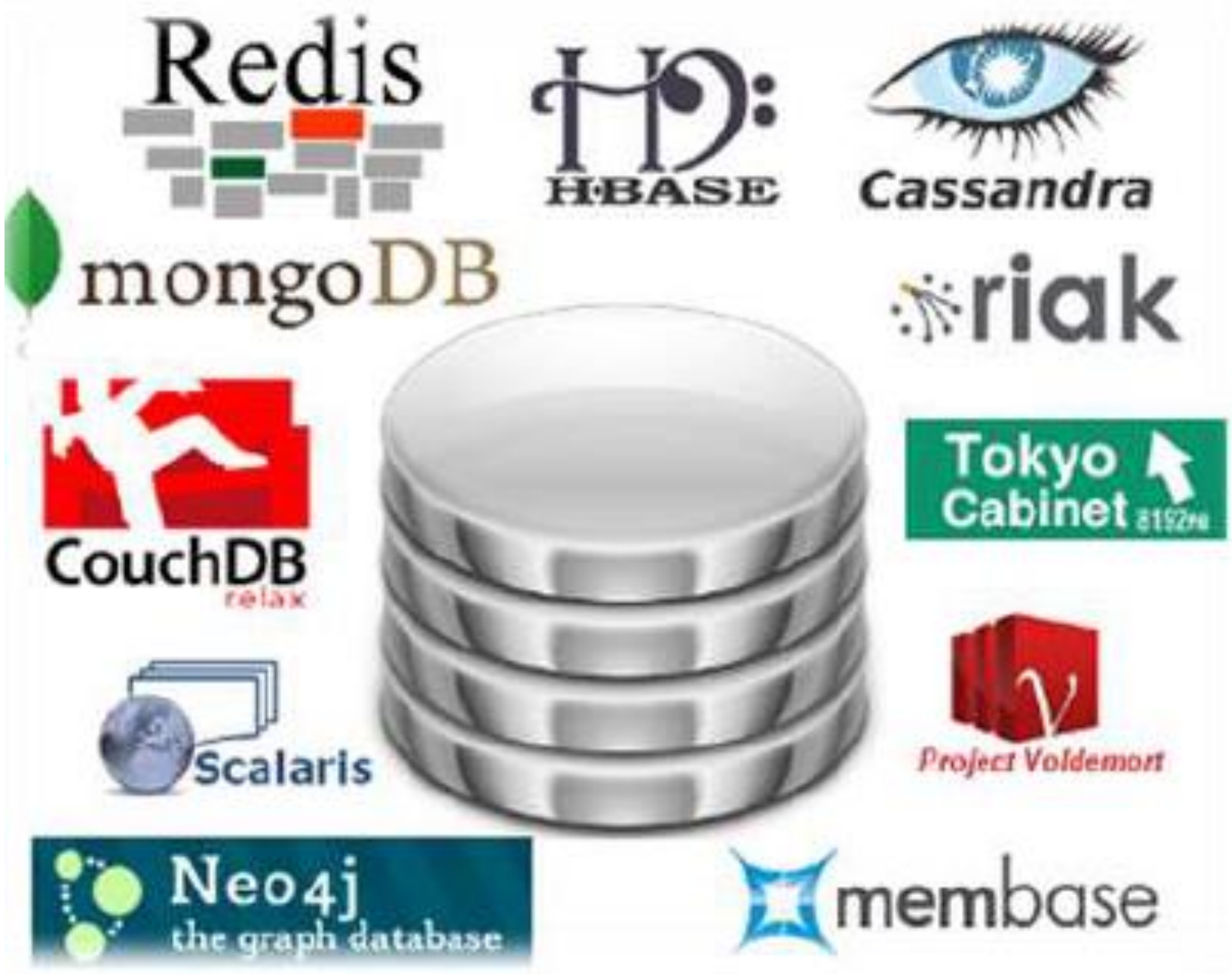
Row/Entity/Record

Relational Database Management Systems (RDMS)

- **Microsoft SQL Server**
- **Oracle**
- **MySQL**
- **Postgre SQL**
- **IBM db2S**

NO SQL DBMS

- **Mongo DB**



Structured query language (SQL)

- **Structured Query Language (SQL)** is a standardized programming language that is used to manage relational databases and perform various operations on the data in them.

- SQL is used for the following:
  - modifying database table and index structures;
  - adding, updating and deleting rows of data; and
  - retrieving subsets of information from within relational database management systems

Tools to write SQL Statements

- **Microsoft:** SQL Server Management studio
- **Oracle:** SQL Developer
- **MySQL:** Workbench
- **Quest Software :** TOAD (with Oracle)

Data Models

- **Data Model:** A set of concepts to describe the structure of a database, and certain constraints that the database should obey.

- **Data Model Operations:** Operations for specifying database retrievals and updates by referring to the concepts of the data model. Operations on the data model may include basic operations and user-defined operations.

Data Models

- **High Level or Conceptual data models** provide concepts that are close to the way many users perceive data, entities, attributes and relationships. (Ex. ERD)

- **Physical data models** describes how data is stored in the computer and the access path needed to access and search for data.

Schemas versus Instances

- **Database Schema:** The description of a database. Includes descriptions of the database structure and the constraints that should hold on the database.
- **Schema Diagram**: A diagrammatic display of (some aspects of) a database schema.
- **Schema Construct**: A component of the schema or an object within the schema.
- **Database Instance:** The actual data stored in a database at a particular moment in time. Also called database state (or occurrence).

Thank You!